



RS003 Week 4 — English Worksheet

Topic: A Three-Part Ship — Body + Nose + Thrusters · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week is all about **thinking about** and **explaining** the three-part ship — a body rectangle, a pointed **nose** triangle, and two **thruster** blocks — and how **draw order** decides what sits in front.

Words to know: **polygon**, **triangle**, **palette**, **draw order**, **thruster**

1 · Predict 🧠

Read each piece of code and write what you think it does. Just read — no running.

```
pygame.draw.rect(screen, CYAN, (CENTER_X - 20, HEIGHT - 80, 40, 50))
```

This is the ship body. Where does it sit on the screen and how big is it?


```
pygame.draw.polygon(screen, WHITE, [(CENTER_X, HEIGHT - 100), (CENTER_X - 20, HEIGHT - 80), (CENTER_X + 20, HEIGHT - 80)])
```

A polygon connects three points. One point is higher than the other two. What shape does it make, and where does it sit on the ship?


```
screen.fill(SPACE)
pygame.draw.rect(screen, CYAN, ...) # body
pygame.draw.polygon(screen, WHITE, ...) # nose
pygame.draw.rect(screen, RED, ...) # thruster
```

Things are drawn from first to last. Which one ends up on top — the space, the body, or the nose?


```
pygame.draw.rect(screen, RED, (CENTER_X - 16, y + 50, 8, 10))    # thruster L  
pygame.draw.rect(screen, RED, (CENTER_X + 8, y + 50, 8, 10))    # thruster R
```

These two lines draw the thrusters. What is different between them, and why does that put one on each side?

2 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Write the fixed code, then explain why the original was wrong.

Bug A — The body should appear **behind** the nose. Right now the body is drawn last and covers the nose.

```
pygame.draw.polygon(screen, WHITE, [(CENTER_X, y - 20), (CENTER_X - 20, y),  
(CENTER_X + 20, y)])  
pygame.draw.rect(screen, CYAN, (CENTER_X - 20, y, 40, 50))
```

Hint: whatever is drawn last sits on top.

Write the fixed code (show both lines in the right order):

Why was it wrong? Why does your fix work?

Bug B — The nose is supposed to be a **triangle** pointing up, but the point is at the bottom and it looks upside down.

```
pygame.draw.polygon(screen, WHITE, [(CENTER_X, y + 20), (CENTER_X - 20, y),  
(CENTER_X + 20, y)])
```

Hint: to point up, the tip point needs a **smaller** Y than the two base points.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The ship should have **two** thrusters, one on each side. Right now both thrusters land on top of each other in the same spot.

```
pygame.draw.rect(screen, RED, (CENTER_X - 16, y + 50, 8, 10))
pygame.draw.rect(screen, RED, (CENTER_X - 16, y + 50, 8, 10))
```

Hint: the second thruster needs a different X so it sits on the other side.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Read this full program. It draws the three-part ship. Answer the questions below by reading — you do not need to run it.

```
import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))
clock = pygame.time.Clock()

SPACE = (10, 12, 40)
CYAN = (80, 220, 255)
WHITE = (255, 255, 255)
RED = (230, 60, 60)

y = HEIGHT - 80

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    screen.fill(SPACE)
    pygame.draw.rect(screen, CYAN, (CENTER_X - 20, y, 40, 50))
    # body
    pygame.draw.polygon(screen, WHITE, [(CENTER_X, y - 20), (CENTER_X - 20, y),
(CENTER_X + 20, y)]) # nose
    pygame.draw.rect(screen, RED, (CENTER_X - 16, y + 50, 8, 10))
    # thruster L
    pygame.draw.rect(screen, RED, (CENTER_X + 8, y + 50, 8, 10))
    # thruster R
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

What do `SPACE` , `CYAN` , `WHITE` , and `RED` store, and why are they written at the top with capital letters?

The line `screen.fill(SPACE)` runs first inside the loop. What does it do before any shapes are drawn?

Look at the three points in the `polygon` line. Which point is the tip of the nose, and how can you tell?

The two thruster lines both use `RED` . What makes one go to the left and the other to the right?

If you moved the body line to be the *last* draw line, what would happen to the nose and thrusters?

4 · Explain Your Lesson Code 🎥

In today's live lesson you wrote your own three-part ship code. Now record a short phone video explaining **the code you wrote**. You can show it running. Try to use these words: **polygon, triangle, palette, draw order, thruster**.

1. Show your code and name the three parts — body, nose, thrusters.
2. Read your polygon line out loud and explain its three points.
3. Explain how draw order decides what sits in front in your ship.
4. Point to your palette colours and say what mood they give.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.